



Laboratory Report # 5

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Date Completed: November 8, 2024

Laboratory Exercise Title: Behavioral Modeling of Combinational Circuits

Target Course Outcomes:

CO1: Create descriptions of digital hardware components such as in combinational and sequential circuits using synthesizable Verilog HDL constructs.

CO2: Verify the functionality of HDL-based components through design verification tools.

Exercise 5A: Hexadecimal Digit to 7-Segment Decoder. Create a behavioral description with a case statement of a hex-to-7-segment decoder. Take note that the target 7-segment output is configured as active low (common anode).

In a common-anode 7 segment, an active low input is required for the segment to display. Below is a 7-segment display with the corresponding label of each segment.

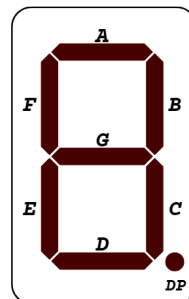


Figure 1. Seven-segment Display.

Below is the appropriate Hexadecimal Digit Patterns to be displayed in the seven-segment display corresponding to the hexadecimal equivalent of the 4-bit binary input.

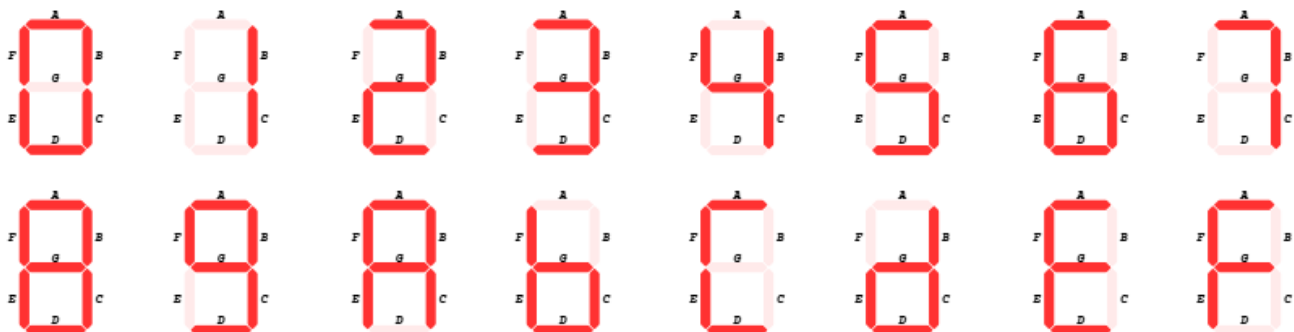


Figure 2. Hexadecimal Digit Patterns.



The hexadecimal digit patterns are tabulated in the following output table below.

Table 1. Output Table of a Common Anode 7-Segment Display

Inputs				Outputs (SSEG)						
w	x	y	z	A	B	C	D	E	F	G
0	0	0	0	0	0	0	0	0	0	1
0	0	0	1	1	0	0	1	1	1	1
0	0	1	0	0	0	1	0	0	1	0
0	0	1	1	0	0	0	0	1	1	0
0	1	0	0	1	0	0	1	1	0	0
0	1	0	1	0	1	0	0	1	0	0
0	1	1	0	0	1	0	0	0	0	0
0	1	1	1	0	0	0	1	1	1	1
1	0	0	0	0	0	0	0	0	0	0
1	0	0	1	0	0	0	0	1	0	0
1	0	1	0	0	0	0	1	0	0	0
1	0	1	1	1	1	0	0	0	0	0
1	1	0	0	0	1	1	0	0	0	1
1	1	0	1	1	0	0	0	0	1	0
1	1	1	0	0	1	1	0	0	0	0
1	1	1	1	0	1	1	1	0	0	0

Table 2. Output Table of the Decimal Point input of the Seven-Segment Display

Input (DP)	Output (DP)
0	1
1	0



```
HexTo7Seg.v
7  * PROJECT DESCRIPTION:
8  *   A Verilog File of a Hexadecimal Digit to 7-Segment
9  *   Conversion Circuit
10 *
11 * *****/
12
13 module HexTo7Seg(
14     input [3:0] Hex,
15     input DP,
16     output reg [7:0] SSeg
17 );
18
19     reg [6:0] dig;
20
21     always @(*)
22     begin
23         case (Hex)
24             4'd0 : dig = 7'b0000001;
25             4'd1 : dig = 7'b1001111;
26             4'd2 : dig = 7'b0010010;
27             4'd3 : dig = 7'b0000110;
28             4'd4 : dig = 7'b1001100;
29             4'd5 : dig = 7'b0100100;
30             4'd6 : dig = 7'b0100000;
31             4'd7 : dig = 7'b0001111;
32             4'd8 : dig = 7'b0000000;
33             4'd9 : dig = 7'b0000100;
34             4'd10 : dig = 7'b0001000;
35             4'd11 : dig = 7'b1100000;
36             4'd12 : dig = 7'b0110001;
37             4'd13 : dig = 7'b1000010;
38             4'd14 : dig = 7'b0110000;
39             4'd15 : dig = 7'b0111000;
40         endcase
41         SSeg = {dig, ~DP};
42     end
43
44 end
45
46 endmodule
47
48
```

Figure 3. Verilog File for A Hexadecimal to 7-Segment Decoder Circuit.

Quartus Prime Lite Edition - Z:/HDL Program Files/Exercise5A/HexTo7Seg - HexTo7Seg

Entity/Instance: MAX 10: 10M02DCU324A6G

Table of Contents: Flow Summary, Flow Settings, Flow Non-Default Global Settings, Flow Elapsed Time, Flow OS Summary, Flow Log, Analysis & Synthesis, Flow Messages, Flow Suppressed Messages

Tasks: Compilation, Task: Compile Design, Analysis & Synthesis, Edit Settings, View Report, Analysis & Elaboration

Flow Summary

Flow Status	Successful - Fri Oct 11 14:06:47 2024
Quartus Prime Version	20.1.1 Build 720 11/11/2020 SJ Lite Edition
Revision Name	HexTo7Seg
Top-level Entity Name	HexTo7Seg
Family	MAX 10
Device	10M02DCU324A6G
Timing Models	Final
Total logic elements	7
Total registers	0
Total pins	13
Total virtual pins	0
Total memory bits	0
Embedded Multiplier 9-bit elements	0
Total PLLs	0
UFM blocks	0
ADC blocks	0

Messages

Type ID Message

- Running quartus prime analysis & synthesis
- Command: quartus_map --read_settings_files=on --write_settings_files=off HexTo7Seg -c HexTo7Seg
- 18236 Number of processors has not been specified which may cause overloading on shared machines. Set the global assignment NUM_PARALLEL_PROCESSORS in your QSF to an appropriate value for best performance.
- 20030 Parallel compilation is enabled and will use 10 of the 10 processors detected
- 12021 Found 1 design units, including 1 entities, in source file hexTo7seg.v
- 12021 Found 1 design units, including 1 entities, in source file tb_hexTo7seg.v
- 12127 Elaborating entity "hexTo7Seg" for the top level hierarchy
- 286030 Timing-driven synthesis is running
- 16010 Generating hard_block partition "hard_block:auto_generated_inst"
- 21057 Implemented 20 device resources after synthesis - the final resource count might be different
- Quartus Prime Analysis & Synthesis was successful. 0 errors, 1 warning

Figure 4. Flow Summary Compilation for HexTo7Seg.v



```
tb_HexTo7Seg.v
HexTo7Seg.v

9
10 *****/
11
12 `timescale 1 ns/1 ps
13 module tb_HexTo7Seg();
14
15     reg    [3:0] Hex;
16     reg          DP;
17
18     wire [7:0] SSeg;
19
20     HexTo7Seg UUT(Hex, DP, SSeg);
21
22     initial begin
23
24         DP = 0 ; Hex = 4'd0; #10
25         Hex = 4'd1; #10
26         Hex = 4'd2; #10
27         Hex = 4'd3; #10
28         Hex = 4'd4; #10
29         Hex = 4'd5; #10
30         Hex = 4'd6; #10
31         Hex = 4'd7; #10
32         Hex = 4'd8; #10
33         Hex = 4'd9; #10
34         Hex = 4'd10; #10
35         Hex = 4'd11; #10
36         Hex = 4'd12; #10
37         Hex = 4'd13; #10
38         Hex = 4'd14; #10
39         Hex = 4'd15; #10
40
41         DP = 1 ; Hex = 4'd0; #10
42         Hex = 4'd1; #10
43         Hex = 4'd2; #10
44         Hex = 4'd3; #10
45         Hex = 4'd4; #10
46         Hex = 4'd5; #10
47         Hex = 4'd6; #10
48         Hex = 4'd7; #10
49         Hex = 4'd8; #10
50         Hex = 4'd9; #10
51         Hex = 4'd10; #10
52         Hex = 4'd11; #10
53         Hex = 4'd12; #10
54         Hex = 4'd13; #10
55         Hex = 4'd14; #10
56         Hex = 4'd15; #10
57
58     $stop;
59     end
60
61 endmodule
62
```

Figure 5. Testbench File for A Hexadecimal to 7-Segment Decoder Circuit.



Table 3. Output Table and its Hexadecimal Equivalent (input DP = 0, output DP = 1)

Outputs (SSEG)								Hex Equivalent
A	B	C	D	E	F	G	DP	
0	0	0	0	0	0	1	1	03
1	0	0	1	1	1	1	1	9F
0	0	1	0	0	1	0	1	25
0	0	0	0	1	1	0	1	0D
1	0	0	1	1	0	0	1	99
0	1	0	0	1	0	0	1	49
0	1	0	0	0	0	0	1	41
0	0	0	1	1	1	1	1	1F
0	0	0	0	0	0	0	1	01
0	0	0	0	1	0	0	1	09
0	0	0	1	0	0	0	1	11
1	1	0	0	0	0	0	1	C1
0	1	1	0	0	0	1	1	63
1	0	0	0	0	1	0	1	85
0	1	1	0	0	0	0	1	61
0	1	1	1	0	0	0	1	71

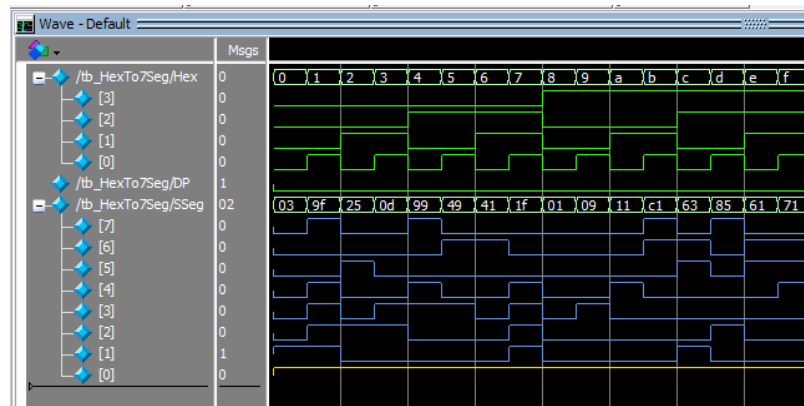


Figure 6. Output waveform for Table 3



Table 4. Output Table and its Hexadecimal Equivalent (input DP = 1, output DP = 0)

Outputs (SSEG)								Hex Equivalent
A	B	C	D	E	F	G	DP	
0	0	0	0	0	0	1	0	02
1	0	0	1	1	1	1	0	9E
0	0	1	0	0	1	0	0	24
0	0	0	0	1	1	0	0	0C
1	0	0	1	1	0	0	0	98
0	1	0	0	1	0	0	0	48
0	1	0	0	0	0	0	0	40
0	0	0	1	1	1	1	0	1E
0	0	0	0	0	0	0	0	00
0	0	0	0	1	0	0	0	08
0	0	0	1	0	0	0	0	10
1	1	0	0	0	0	0	0	C0
0	1	1	0	0	0	1	0	62
1	0	0	0	0	1	0	0	84
0	1	1	0	0	0	0	0	60
0	1	1	1	0	0	0	0	70

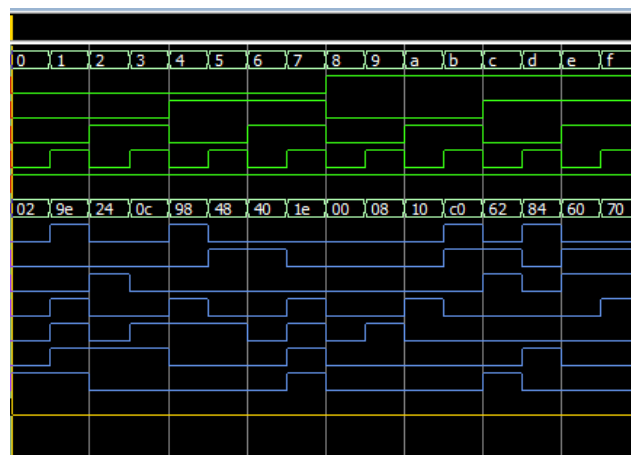


Figure 6. Output waveform for Table 4

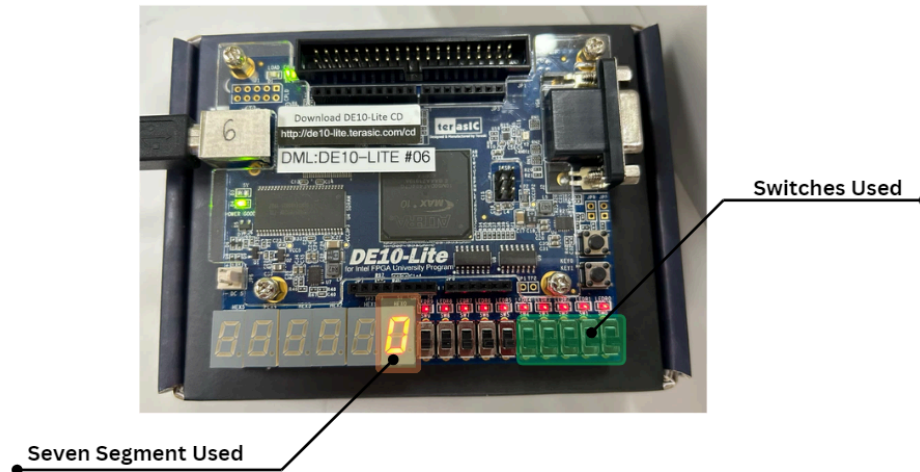


Figure 7. Programmed FPGA switches and seven-segment displays for Activity 5A

Table 5. I/O Assignments for Activity 5A

Verilog Name	FPGA Pin Assignment	Corresponding I/O Device	Description
HEX[0]	PIN_C10	SW0	4-bit Input with SW3 as MSb and SW0 as LSb
HEX[1]	PIN_C11	SW1	
HEX[2]	PIN_D12	SW2	
HEX[3]	PIN_C12	SW3	
DP	PIN_A12	SW4	Decimal Point Switch
SSEG[7]	PIN_C14	HEX0	Seven Segment A-pin
SSEG[6]	PIN_E15		Seven Segment B-pin
SSEG[5]	PIN_C15		Seven Segment C-pin
SSEG[4]	PIN_C16		Seven Segment D-pin
SSEG[3]	PIN_E16		Seven Segment E-pin
SSEG[2]	PIN_D17		Seven Segment F-pin
SSEG[1]	PIN_C17		Seven Segment G-pin
SSEG[0]	PIN_D15		Seven Segment DP-pin



FPGA Outputs:



Figure 8. FPGA Outputs for Activity 5A from 0 to F. (SW4 = 0)

As seen in Figure 8, the highlighted green switches indicate the 4-bit data to be converted into its hexadecimal equivalent. This is then translated into the seven-segment display. The switch highlighted in yellow indicate the switch that enables the decimal point. Since the switch is turned off, the decimal point in the seven segment of all outputs are disabled.

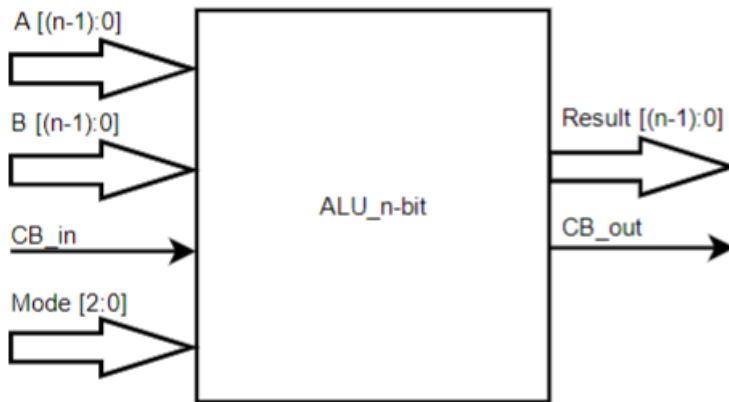


Figure 9. FPGA selected outputs for decimal point enabling.

In figure 9, only selected outputs are displayed since the switch to enable the decimal point is separate from the 4-bit data. We can see that the switch highlighted in yellow for each output (specifically SW4) is enabled.



Exercise 5B: n-Bit Arithmetic and Logic Unit (ALU). Design a behavioral description of an n-bit ALU, using either if-else or case statements. Use a parameter to change the value of n (use n = 4 as the default value).



*CB_out is for the ADD (carry) and SUBTRACT (borrow) modes only; otherwise it should be 0.

Function Table:

Mode	Operation
000	Addition (A + B) with carry
001	Subtract (A – B) with borrow
010	Bitwise AND of A and B
011	Bitwise OR of A and B
100	Bitwise XOR of A and B
101	Complement of A
110	Increment A
111	Decrement A

Figure 10. Block Diagram and Function Table for the n-BIT ALU

```

module Nbit_ALU
#(
    parameter N = 3
)
(
    input      [N-1:0] A, B,
    input      [2:0]   mode,
    output reg [N-1:0] R,
    output reg         CB_out
);

reg [N:0] math;
always @(*)
if (mode == 0) begin
    math = A + B + CB_in ;
    R = math[N-1:0] ;
    CB_out = math[N] ;
end

else if (mode == 1) begin
    math = A - B - CB_in ;
    R = math[N-1:0] ;
    CB_out = math[N] ;
end

else if (mode == 2) begin
    math = 0 ;
    CB_out = 0 ;
    R = A & B ;
end

else if (mode == 3) begin
    math = 0 ;
    CB_out = 0 ;
    R = A | B ;
end

else if (mode == 4) begin
    math = 0 ;
    CB_out = 0 ;
    R = A ^ B ;
end

else if (mode == 5) begin
    math = 0 ;
    CB_out = 0 ;
    R = ~A ;
end

else if (mode == 6) begin
    math = 0 ;
    CB_out = 0 ;
    R = A + 1'b1 ;
end

else begin
    math = 0 ;
    CB_out = 0 ;
    R = A - 1'b1 ;
end

endmodule

```

Figure 11. Verilog File for the n-bit ALU Circuit.

Rather than having N = 4, the HDL code was verified so that it is easily programmable in the FPGA.

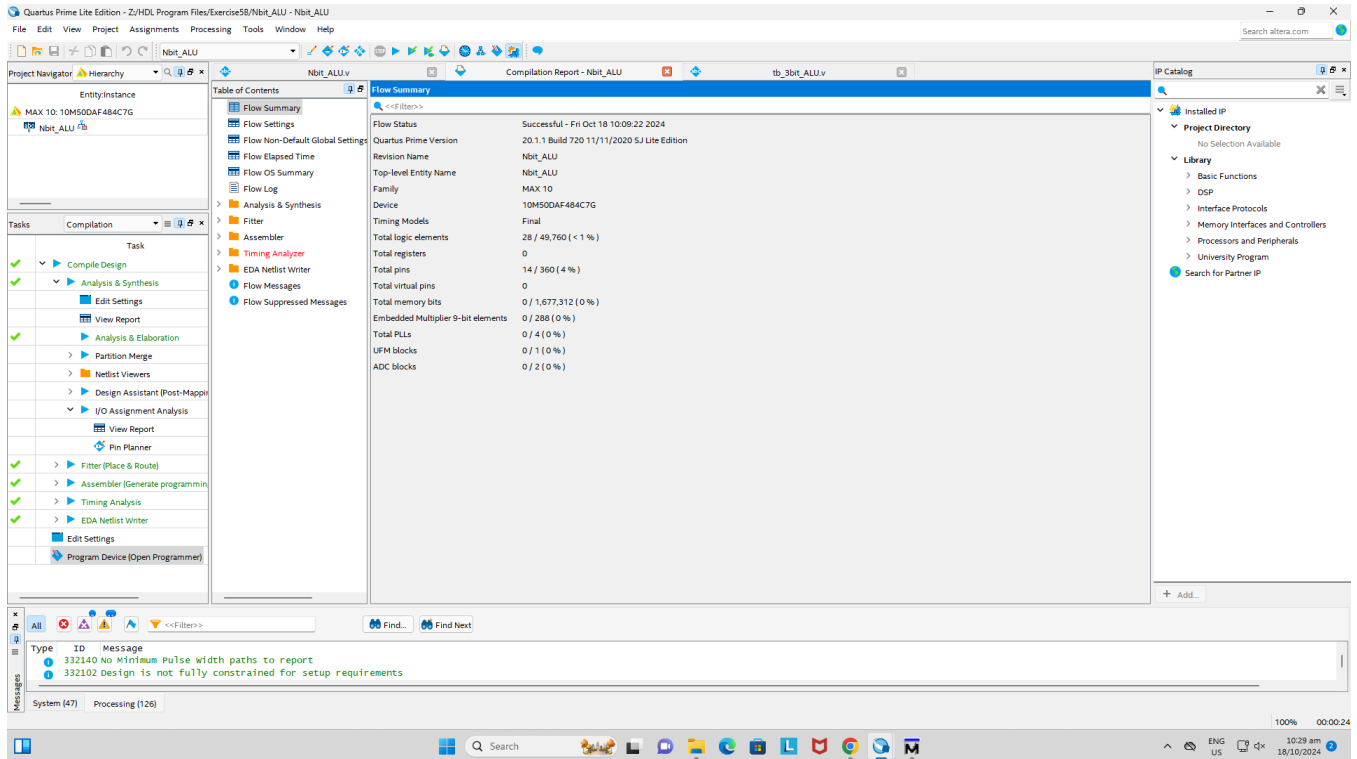


Figure 12. Flow Summary Compilation for HexTo7Seg.v

```

Nbit_ALU #(.N(3))
    UUT(A, B, CB_in, mode, R, CB_out);

initial begin
    mode = 3'd0 ;
    A = 3'd5 ; B = 3'd2 ; CB_in = 0; #10
    A = 3'd3 ; B = 3'd1 ; CB_in = 1; #10
    A = 3'd6 ; B = 3'd7 ; CB_in = 1; #10

    mode = 3'd1 ;
    A = 3'd5 ; B = 3'd2 ; CB_in = 0; #10
    A = 3'd3 ; B = 3'd1 ; CB_in = 1; #10
    A = 3'd6 ; B = 3'd7 ; CB_in = 1; #10

    mode = 3'd2 ;
    A = 3'd5 ; B = 3'd2 ; CB_in = 0; #10
    A = 3'd3 ; B = 3'd1 ; CB_in = 1; #10
    A = 3'd6 ; B = 3'd7 ; CB_in = 1; #10

    mode = 3'd3 ;
    A = 3'd5 ; B = 3'd2 ; CB_in = 0; #10
    A = 3'd3 ; B = 3'd1 ; CB_in = 1; #10
    A = 3'd6 ; B = 3'd7 ; CB_in = 1; #10

    mode = 3'd4 ;
    A = 3'd5 ; B = 3'd2 ; CB_in = 0; #10
    A = 3'd3 ; B = 3'd1 ; CB_in = 1; #10
    A = 3'd6 ; B = 3'd7 ; CB_in = 1; #10

    mode = 3'd5 ;
    A = 3'd5 ; B = 3'd2 ; CB_in = 0; #10
    A = 3'd3 ; B = 3'd1 ; CB_in = 1; #10
    A = 3'd6 ; B = 3'd7 ; CB_in = 1; #10

    mode = 3'd6 ;
    A = 3'd5 ; B = 3'd2 ; CB_in = 0; #10
    A = 3'd3 ; B = 3'd1 ; CB_in = 1; #10
    A = 3'd6 ; B = 3'd7 ; CB_in = 1; #10

    mode = 3'd7 ;
    A = 3'd5 ; B = 3'd2 ; CB_in = 0; #10
    A = 3'd3 ; B = 3'd1 ; CB_in = 1; #10
    A = 3'd6 ; B = 3'd7 ; CB_in = 1; #10

    $stop;
end

```

Figure 13. Testbench Verilog File for n-bit ALU

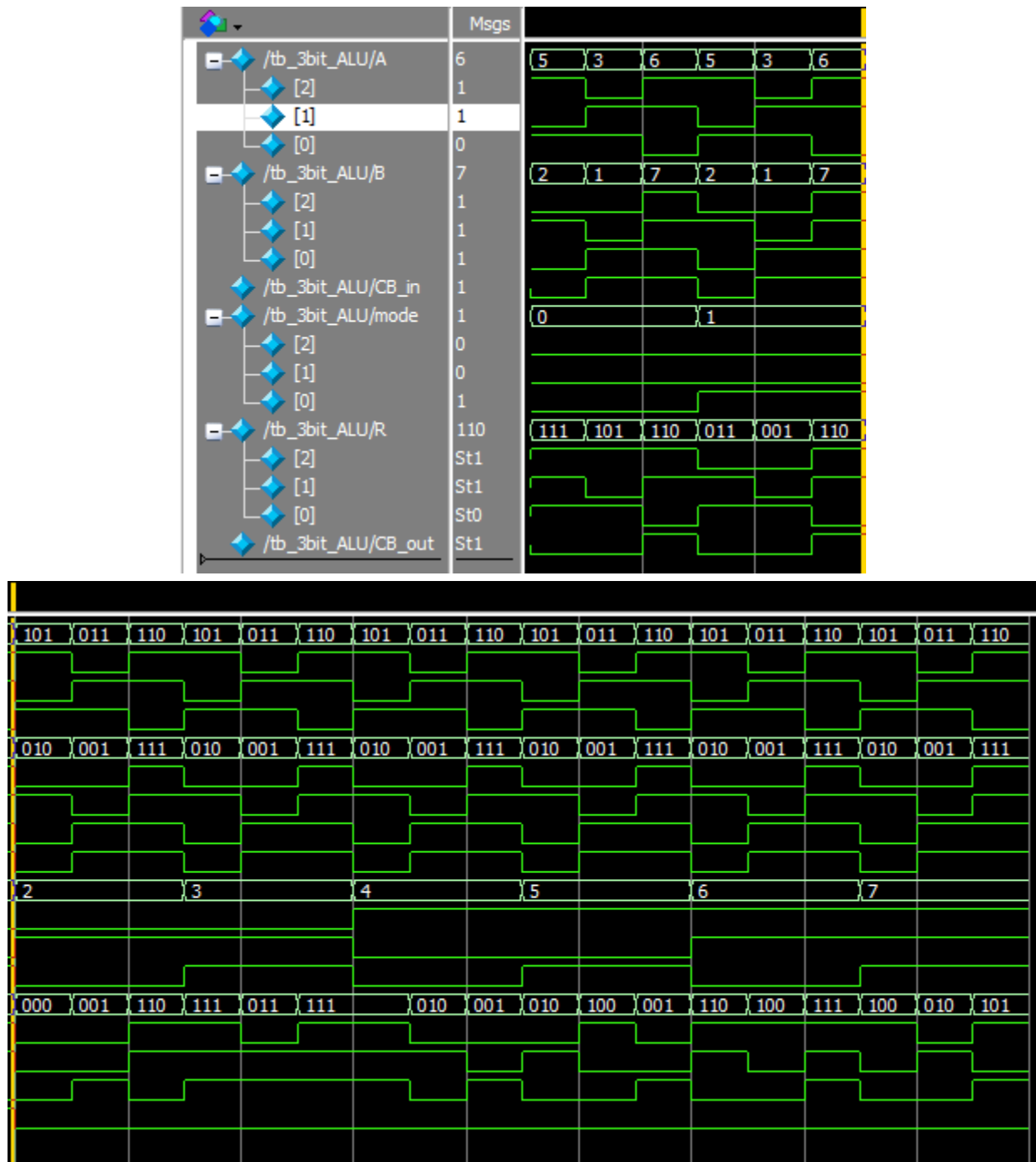


Figure 14. Output wavefore for 5B Testbench File

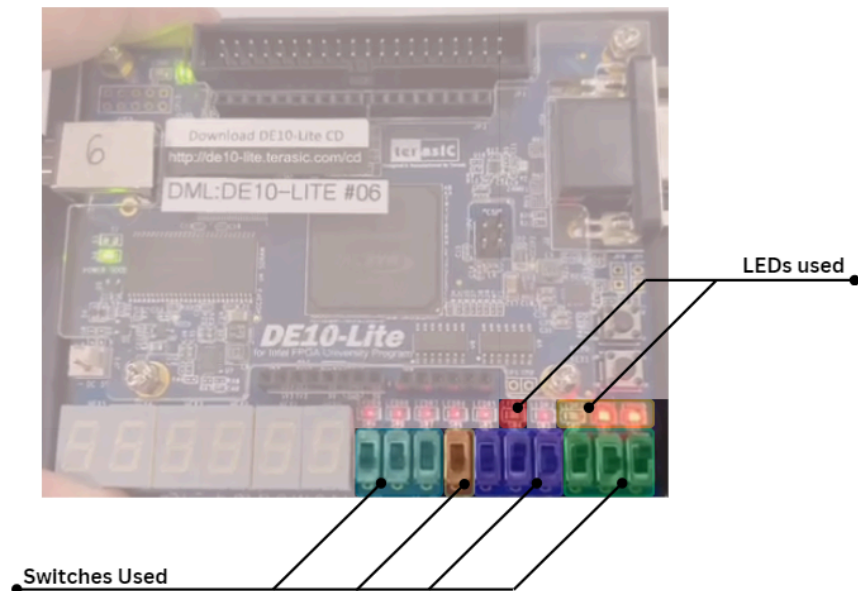


Figure 15. Programmed FPGA switches and seven-segment displays for Activity 5B

Table 5. I/O Assignments for Activity 5B

Verilog Name	FPGA Pin Assignment	Corresponding I/O Device	Description
A[0]	PIN_C10	SW0	3-bit data input (1st Number) [Green]
A[1]	PIN_C11	SW1	
A[2]	PIN_D12	SW2	
B[0]	PIN_C12	SW3	3-bit data input (1st Number) [Blue]
B[1]	PIN_A12	SW4	
B[2]	PIN_B12	SW5	
CB_in	PIN_A13	SW6	Carry/Borrow Input [Orange]
mode[0]	PIN_A14	SW7	mode input [Cyan]
mode[1]	PIN_B14	SW8	
mode[2]	PIN_F15	SW9	
R[0]	PIN_A8	LEDR0	3-bit data output [Yellow]
R[1]	PIN_A9	LEDR1	
R[2]	PIN_A10	LEDR2	
CB_out	PIN_D13	LEDR4	Carry/Borrow Output [Red]

FPGA Outputs:



Figure 16. 3-bit ALU (Modes 0-7[Cyan] with Inputs A[Green] = 5, B[Blue] = 2, CB_in[Orange] = 0)

Based on the modes, varying outputs can be displayed considering that the values for A, B and CB_in are the same. The table below provides a summary into the results and the calculations .

Table 6. ALU Results for A = 5, B = 2 and CB_in = 0

ALU Mode	Description	Calculation	Result
0	Addition Mode (A+B+CB_in)	$5 + 2 + 0 = 7$	111B = 7D
1	Subtraction Mode (A-B+CB_in)	$5 - 2 - 0 = 3$	011B = 3D
2	AND operation (A & B)	$101B \& 010B = 000B$	000B = 0D
3	OR operation (A B)	$101B 010B = 111B$	111B = 7D
4	XOR operation (A ⊕ B)	$101B \oplus 010B = 111B$	111B = 7D
5	Negate A (~A)	$\sim 101B = 010B$	010B = 2B
6	Increment A (A + 1)	$5 + 1 = 6$	110B = 6
7	Decrement A (A - 1)	$5 - 1 = 4$	100B = 4